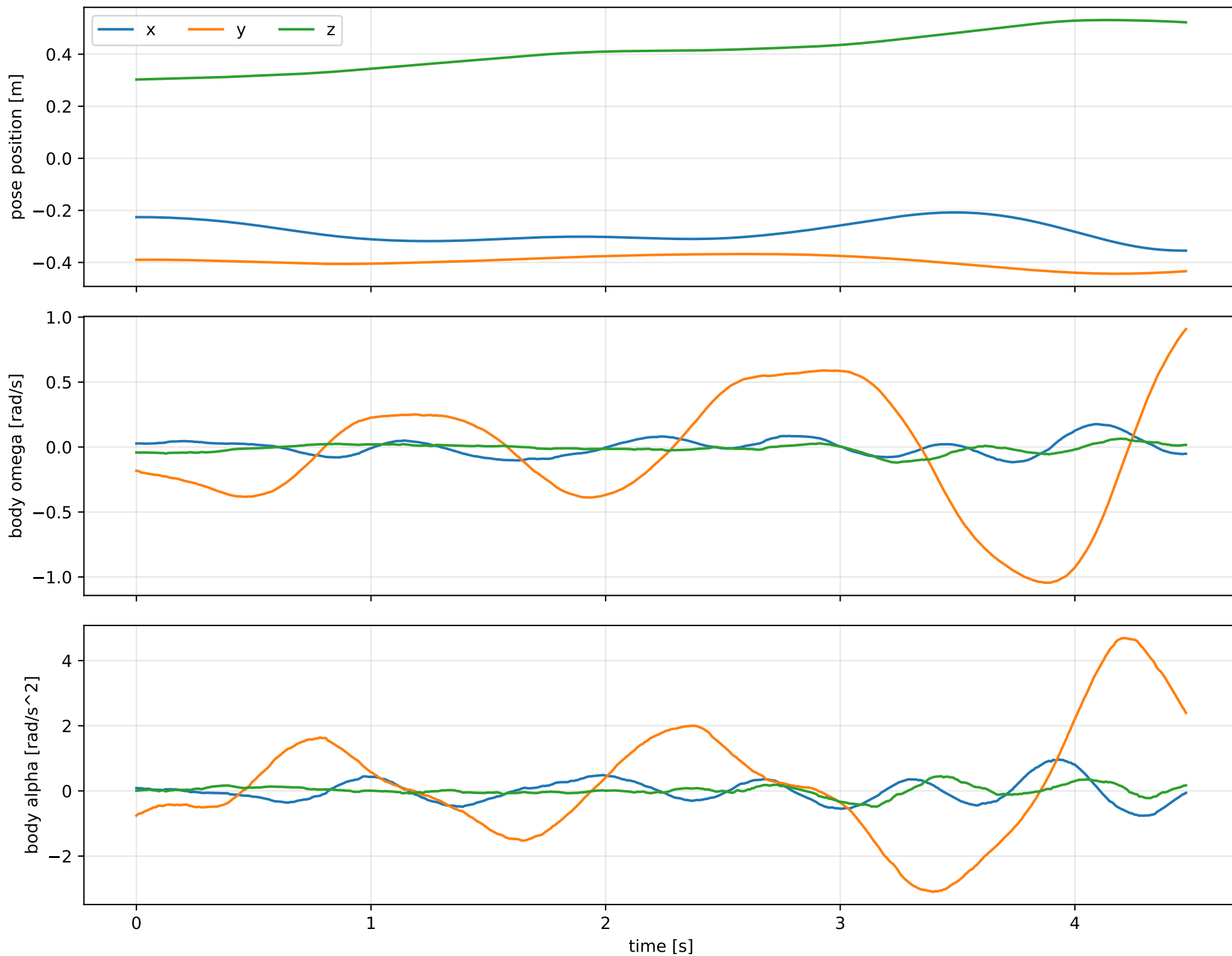
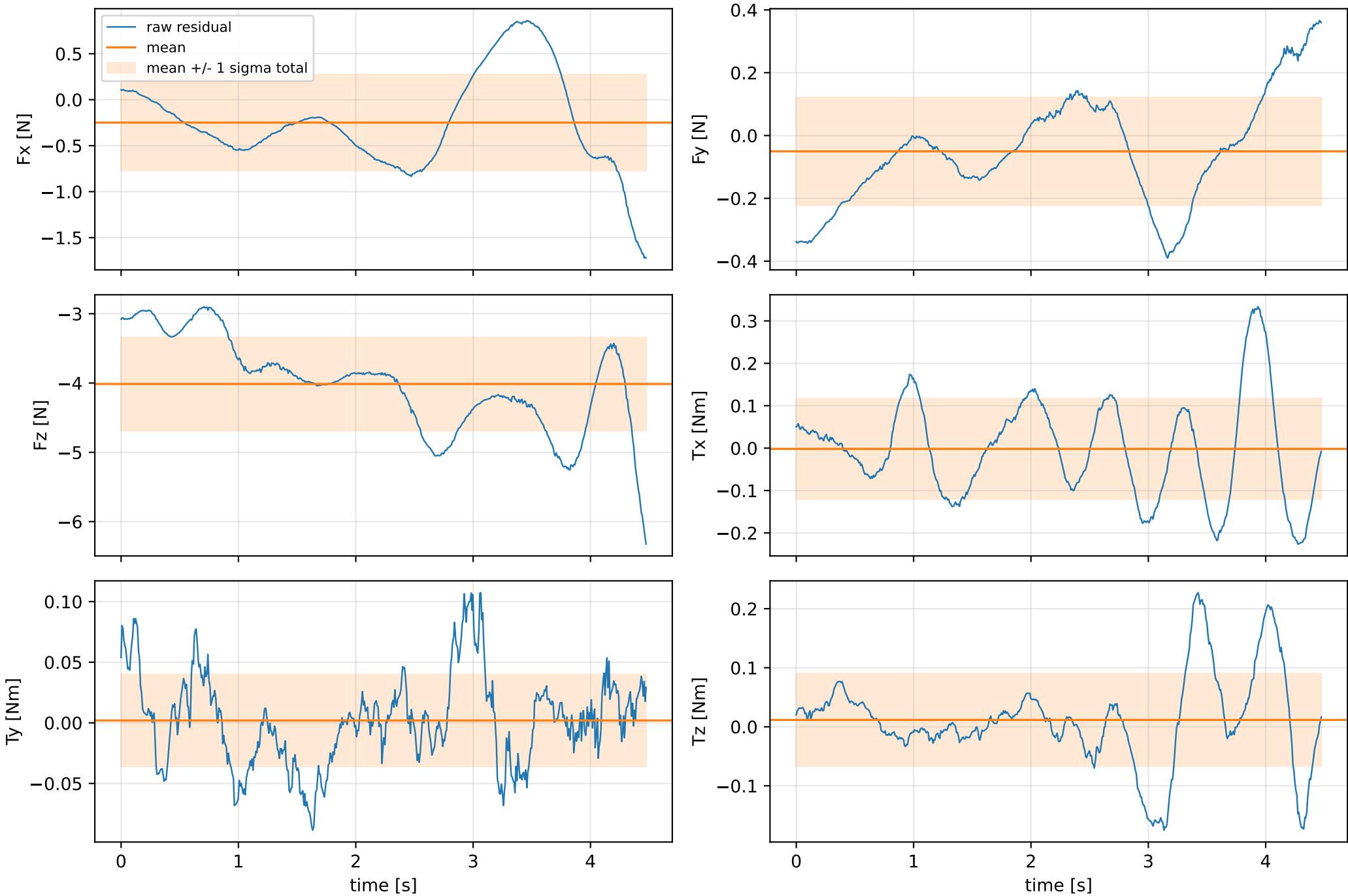


force_over_mass_all_nominal: SG trajectory and derived motion



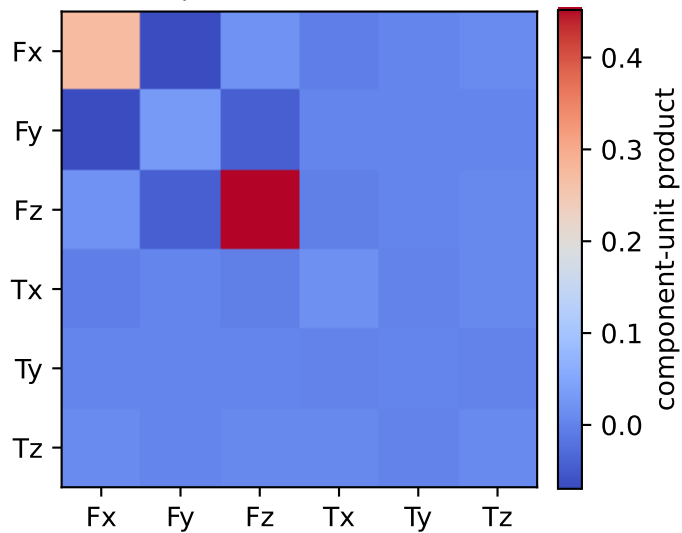
force_over_mass_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



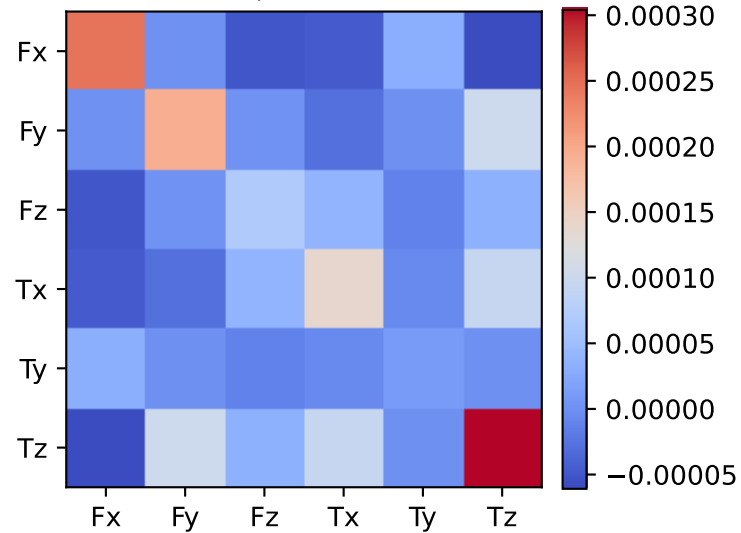
vector RMS about zero: force 4.1146 N, torque 0.14722 Nm; component std about mean: force [0.5209 0.1717 0.6722], torque [0.1183 0.0379 0.0783]

force_over_mass_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.009 0.016 0.285 0.461]

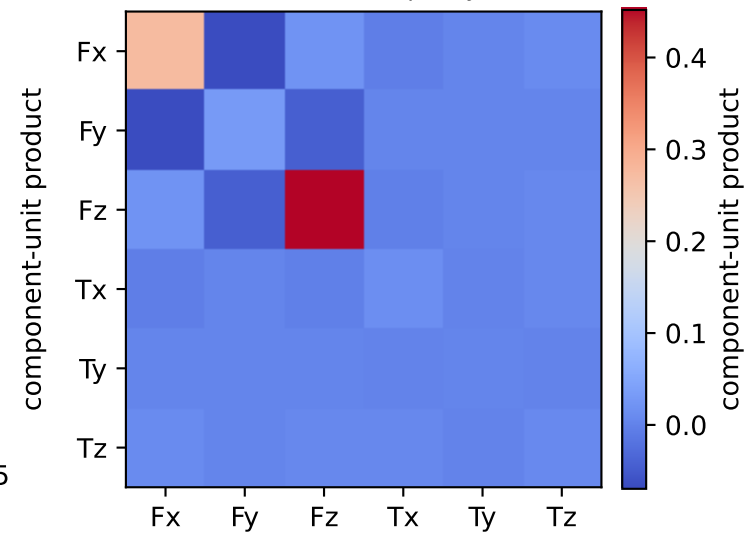
empirical total covariance



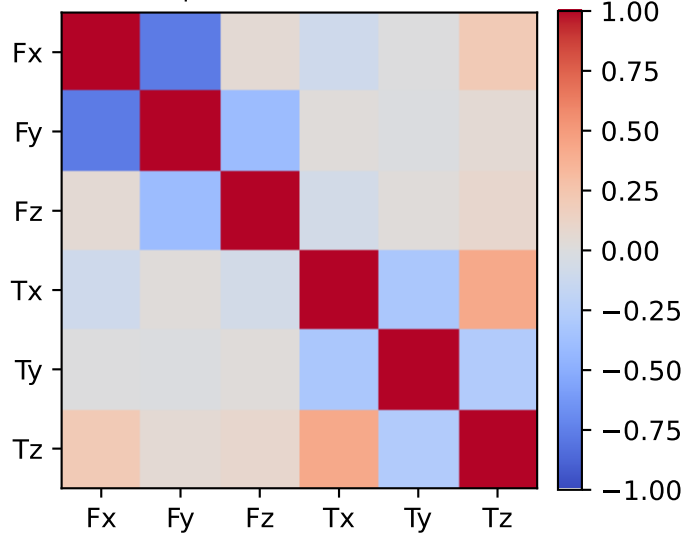
mean full-SG prediction covariance



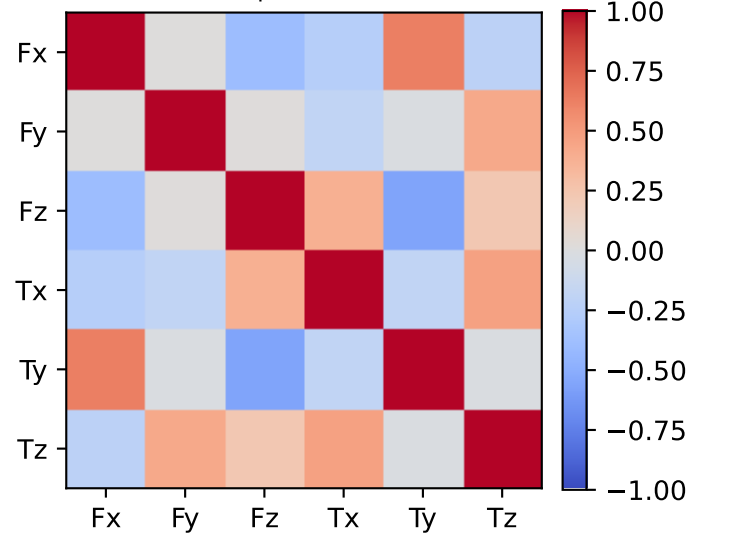
PSD excess model discrepancy covariance



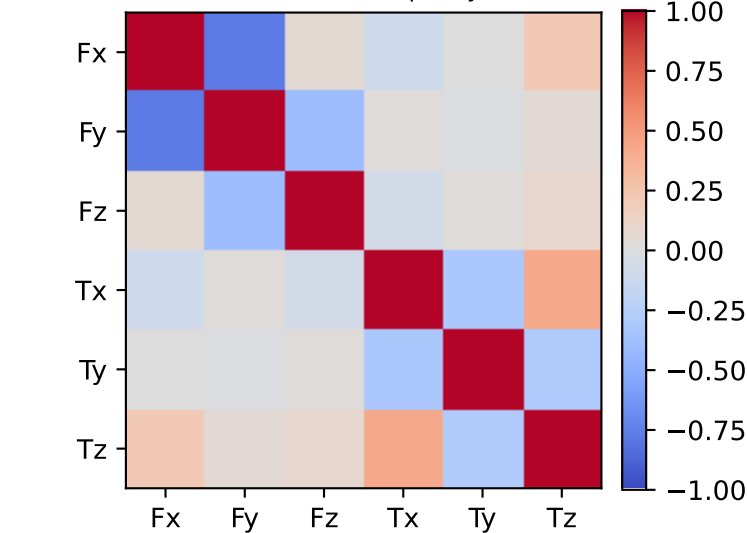
empirical total correlation



mean full-SG prediction correlation

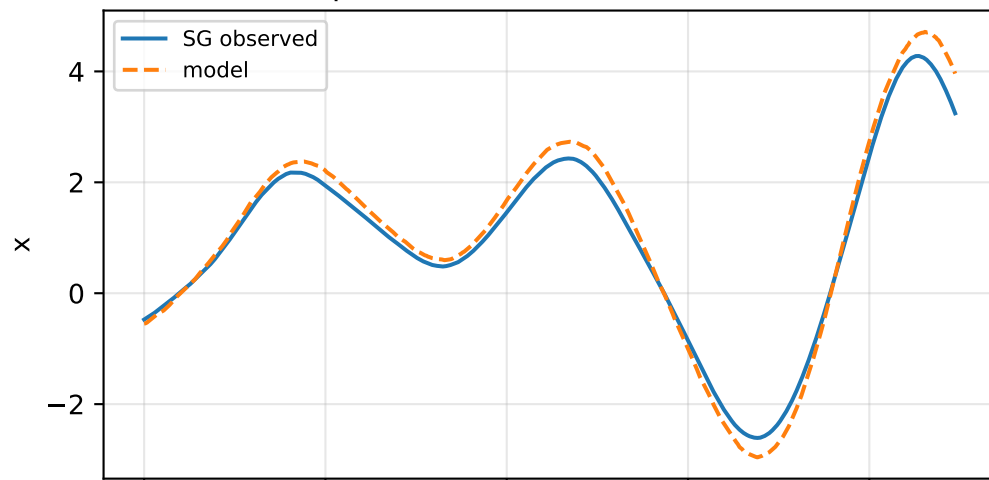


PSD excess model discrepancy correlation

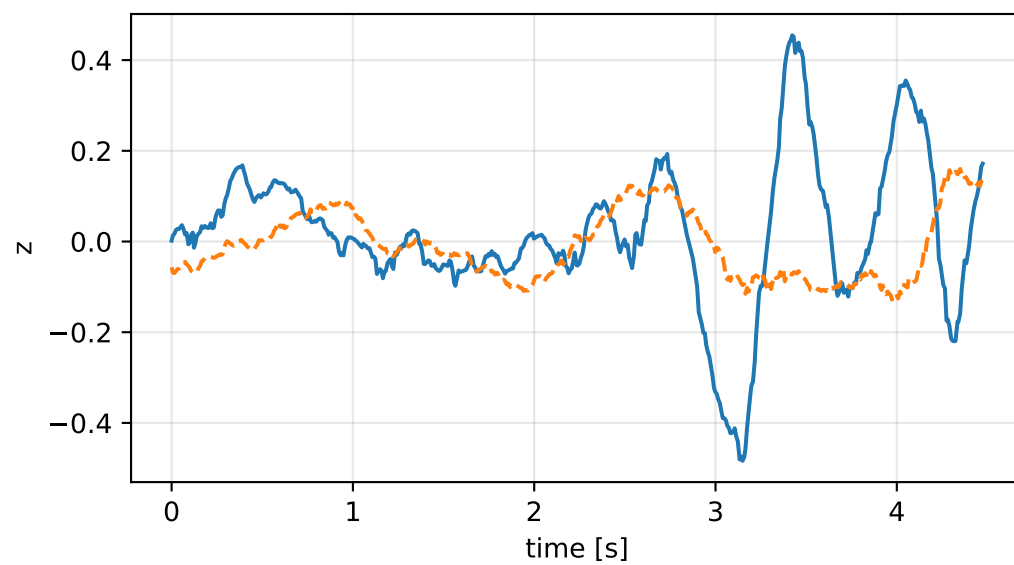
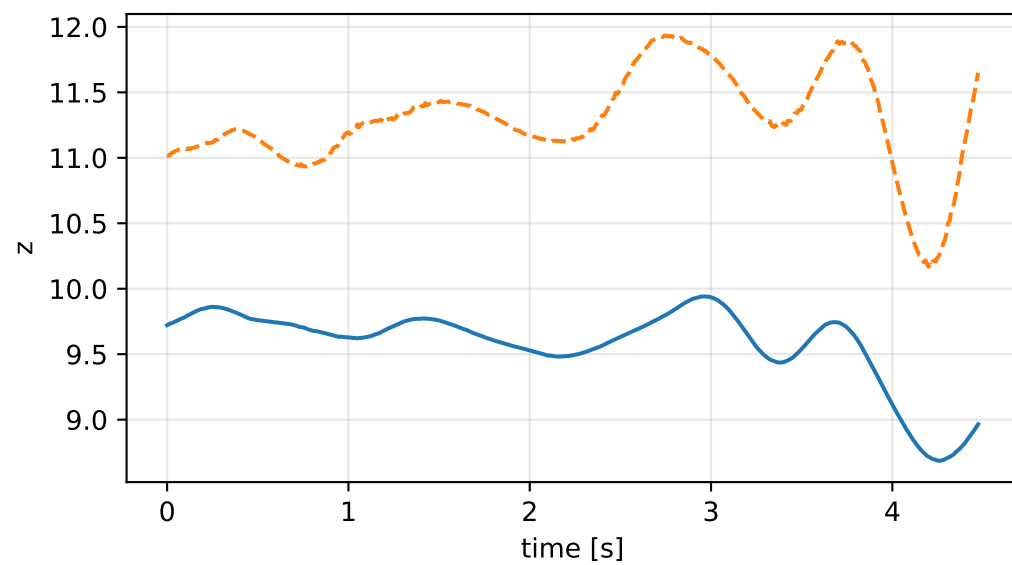
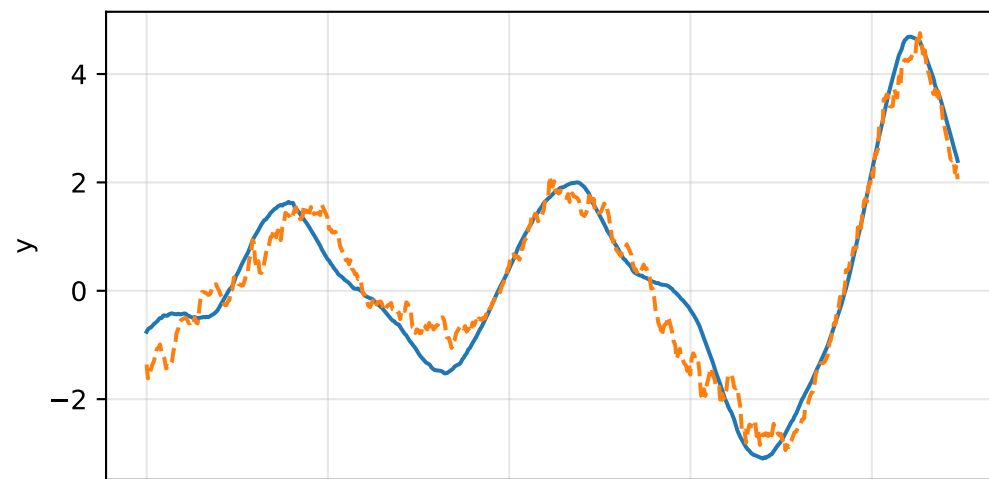
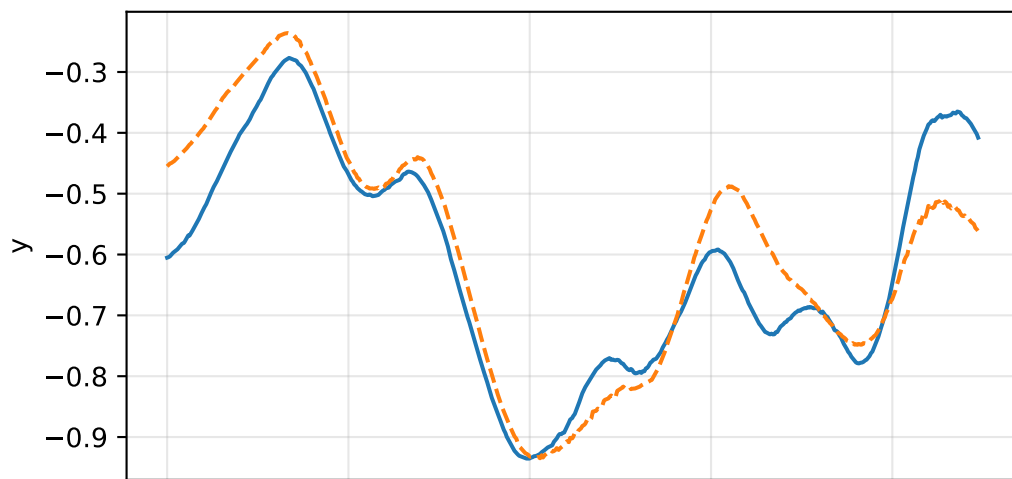
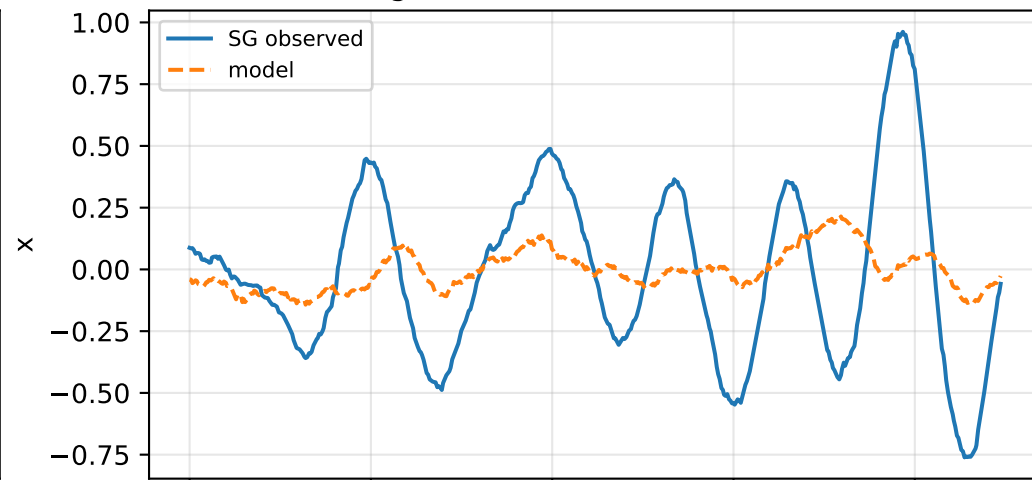


force_over_mass_all_nominal: acceleration residual objective

specific acceleration [m/s^2]

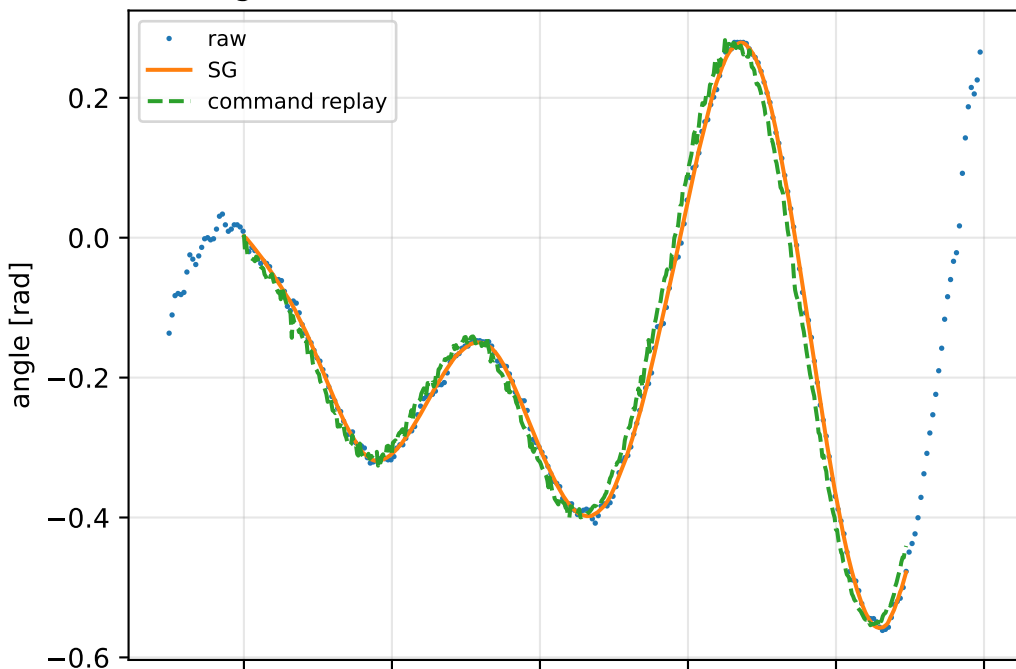


angular acceleration [rad/s^2]

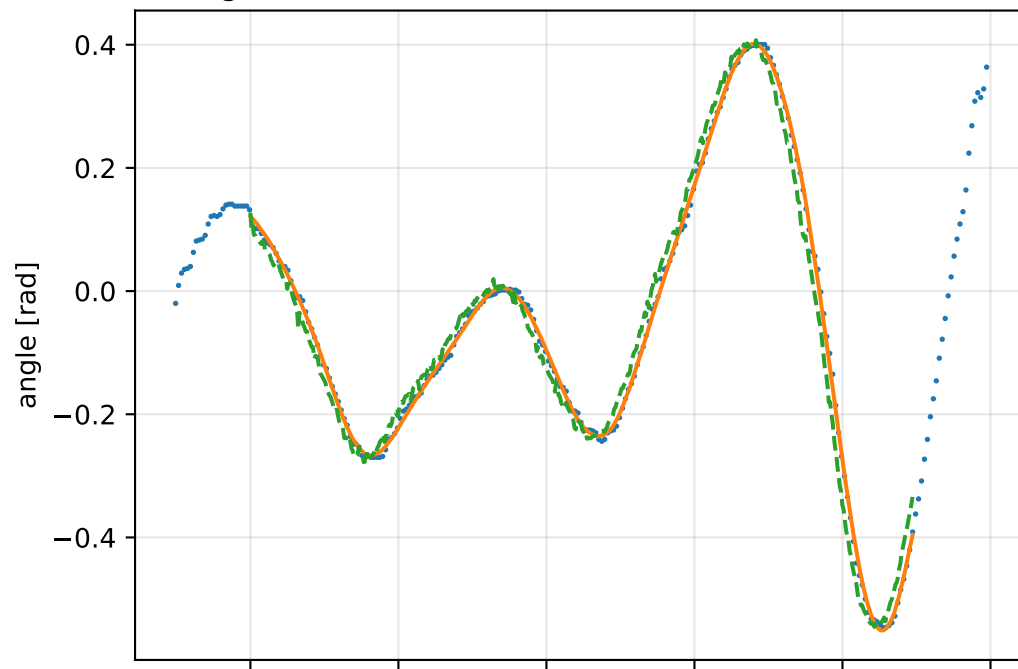


force_over_mass_all_nominal: gimbal measurement audit (objective=measured_sg)

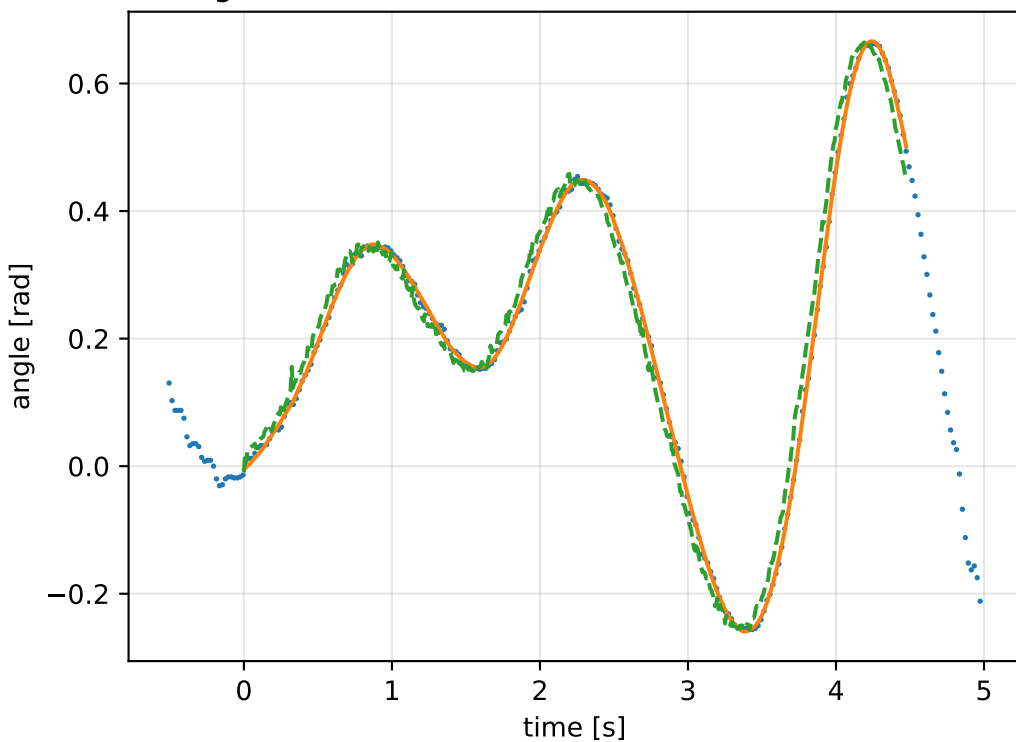
gimbal 1: RMSE=0.027, max=0.07529 rad



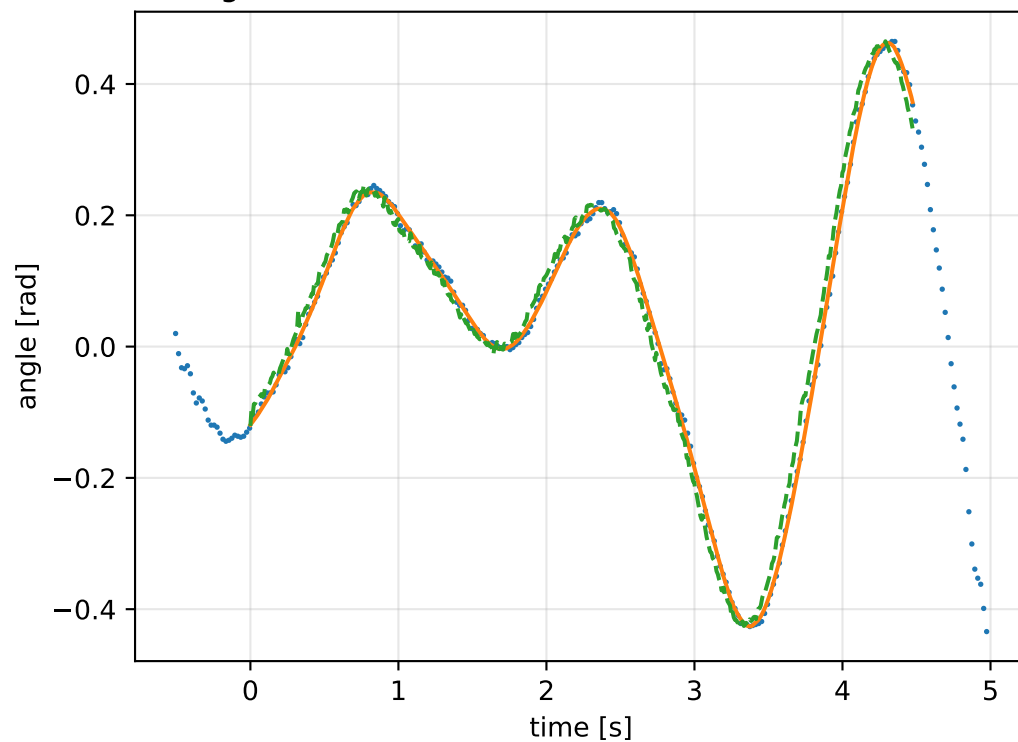
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

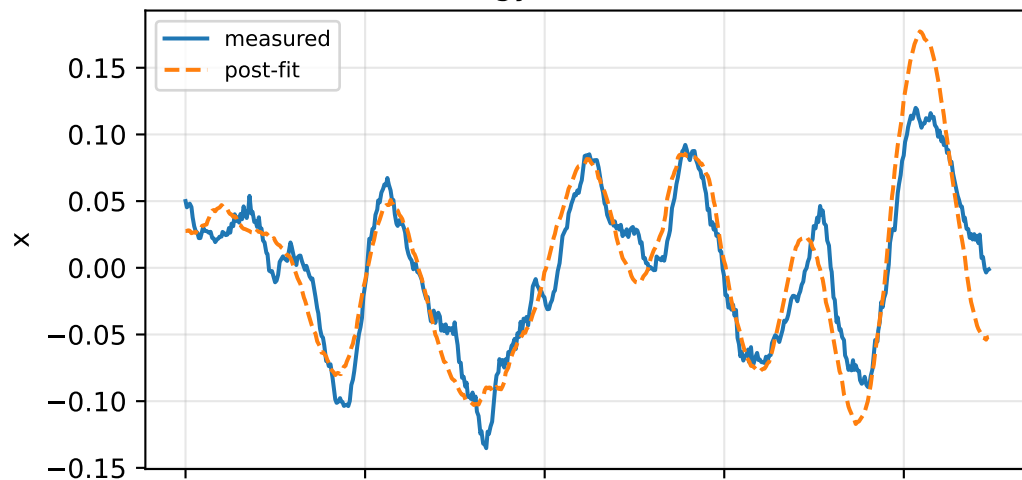


gimbal 4: RMSE=0.02598, max=0.07261 rad

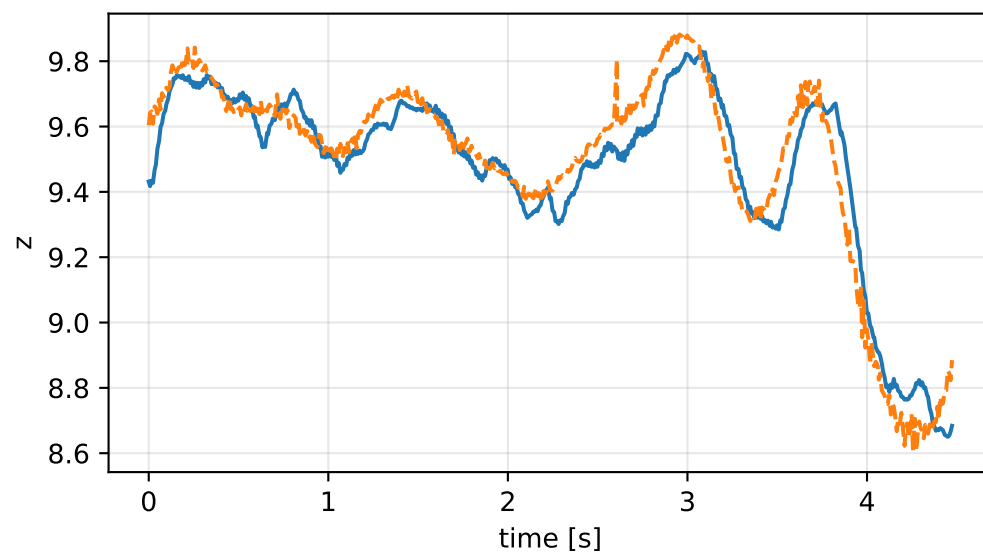
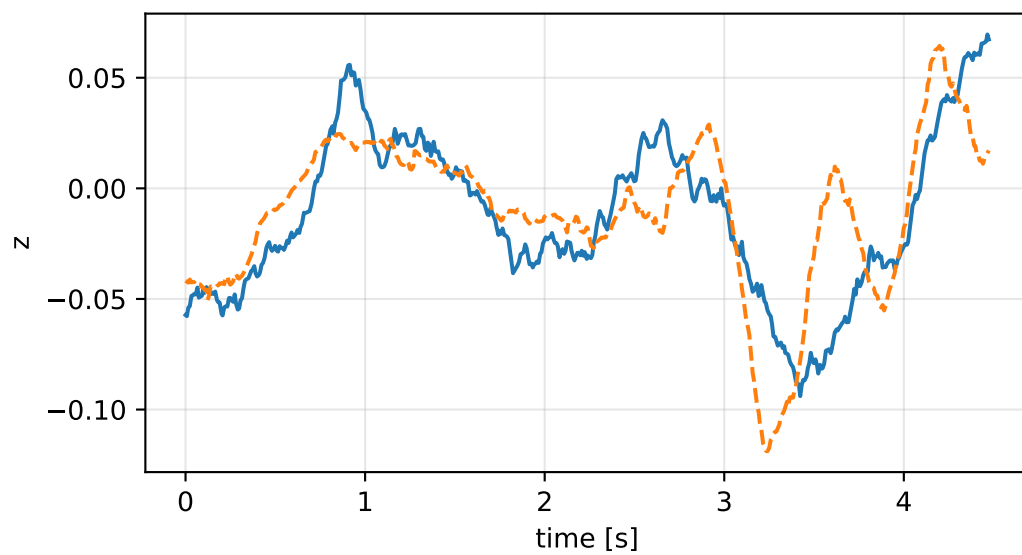
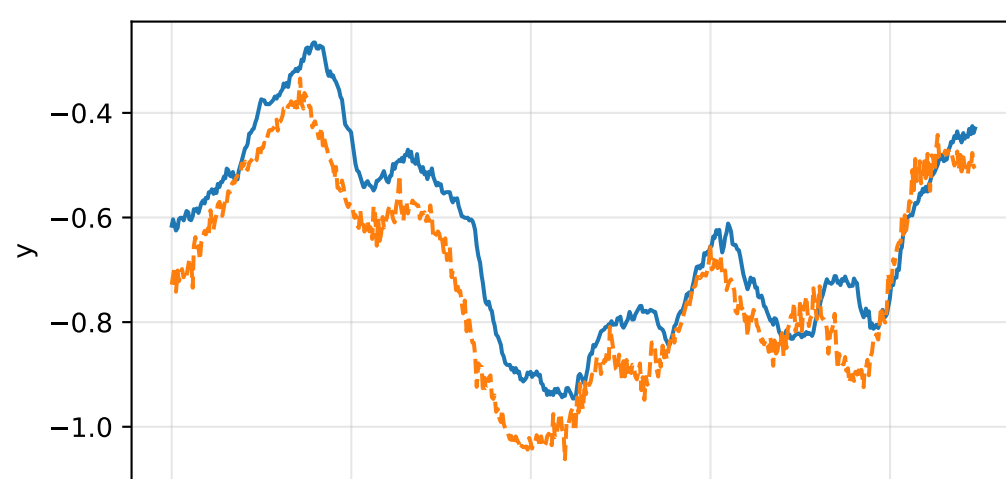
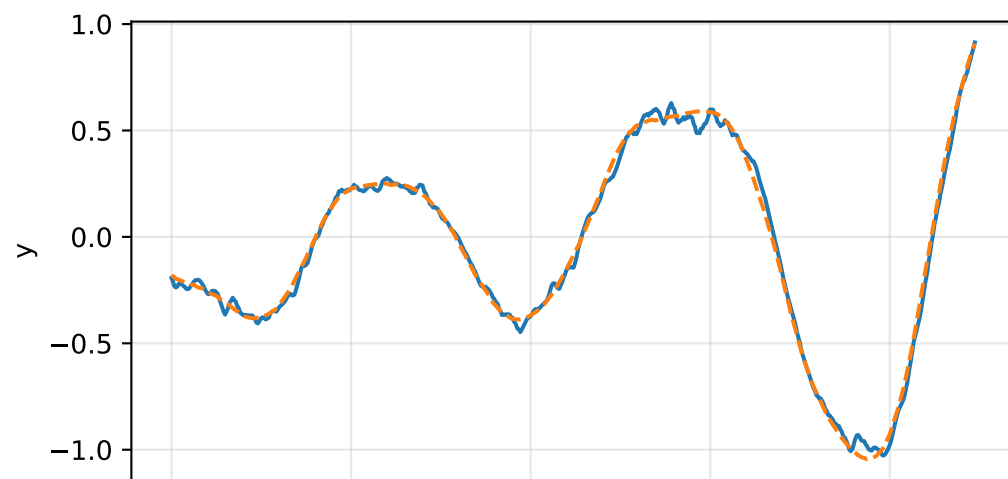
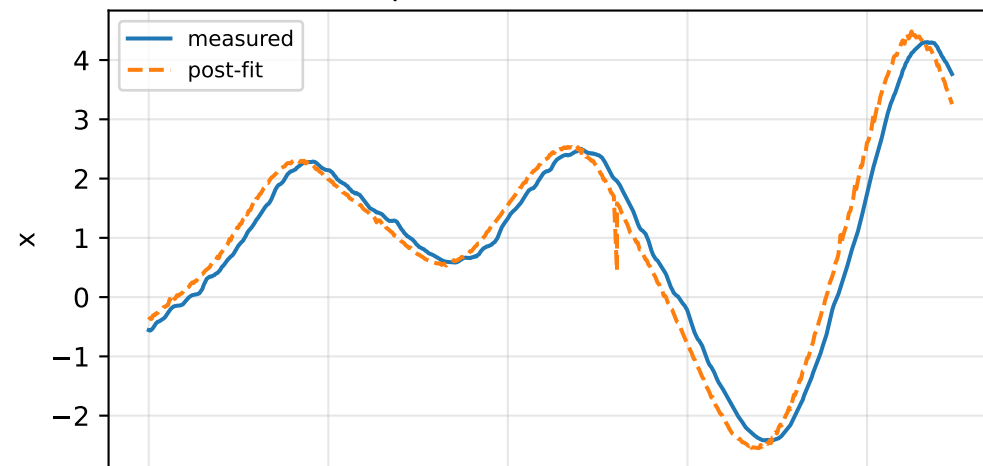


force_over_mass_all_nominal: IMU comparison (diagnostic only)

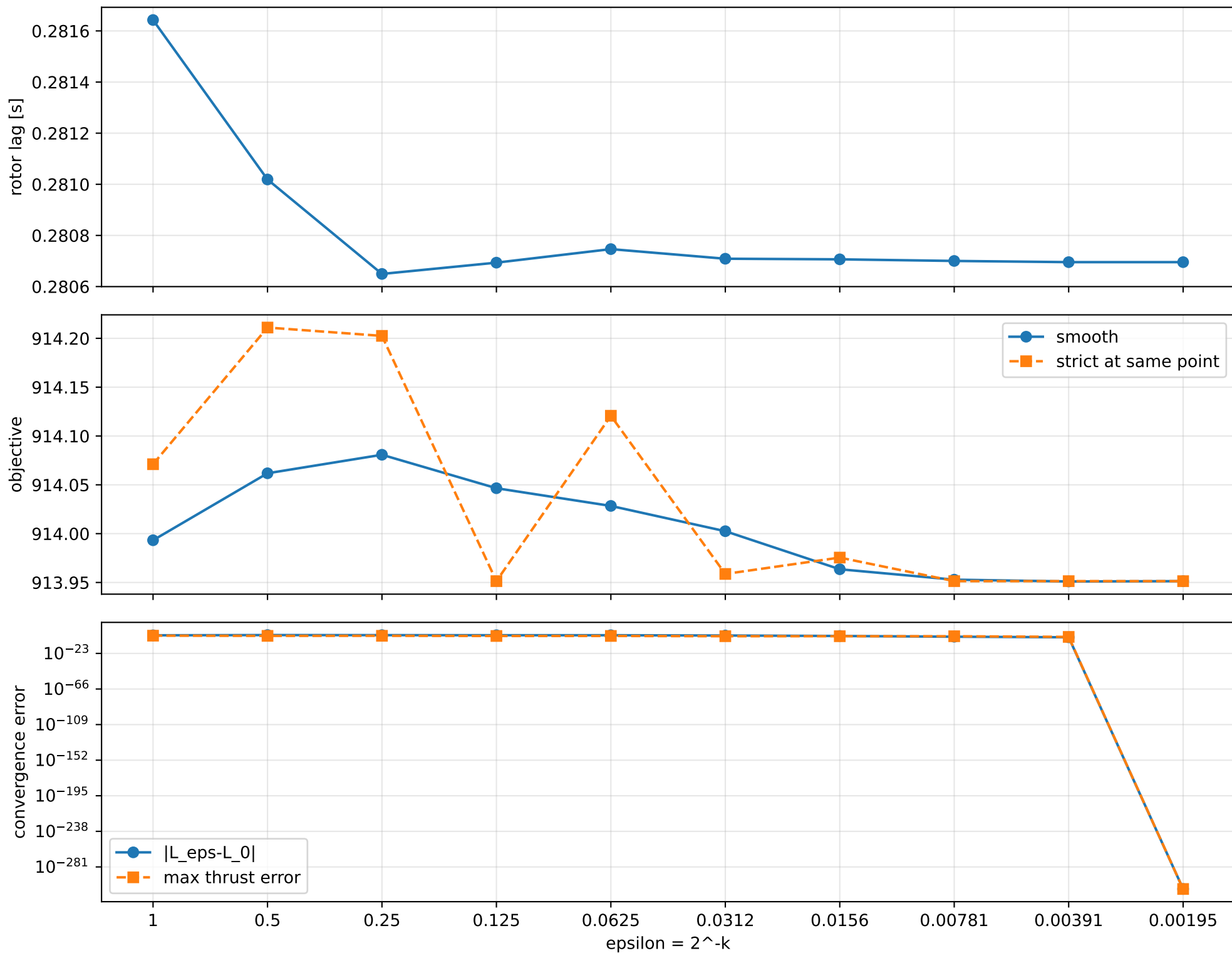
gyro [rad/s]



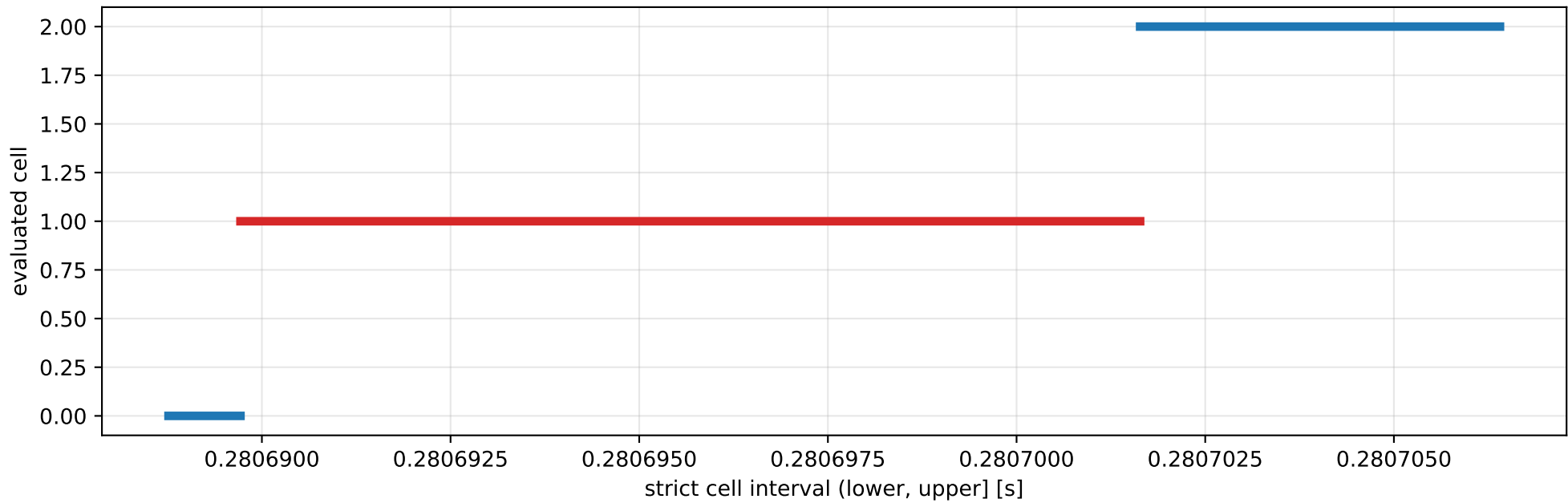
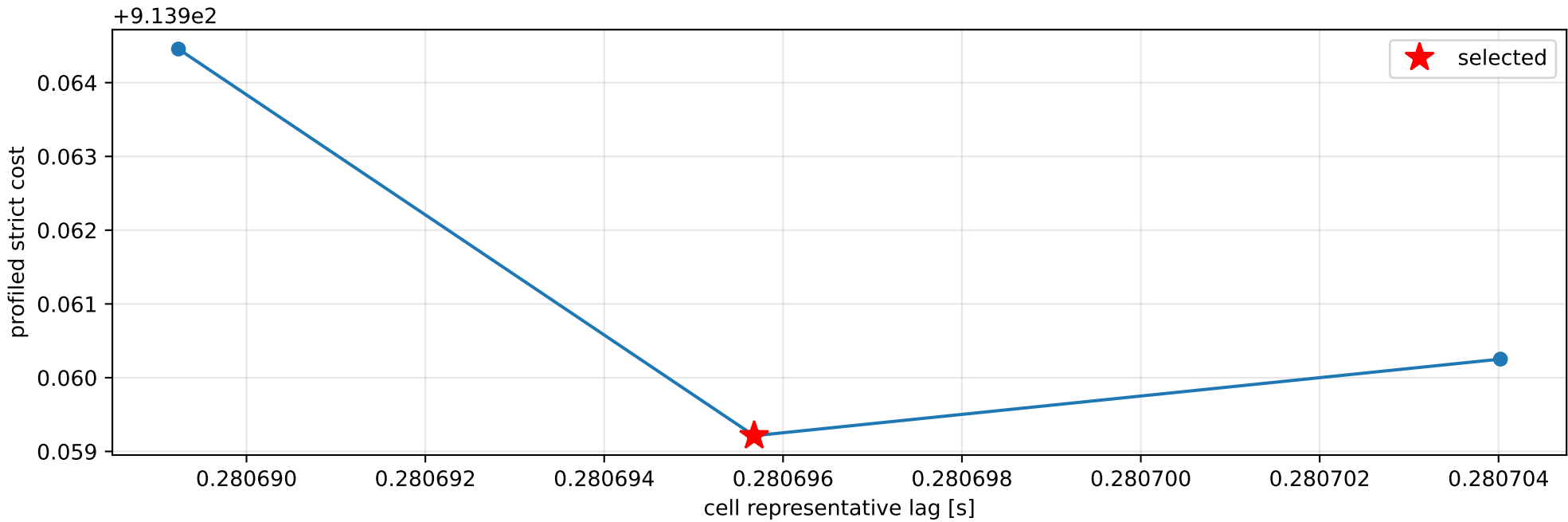
specific force [m/s^2]



force_over_mass_all_nominal: smooth-to-strict continuation



force_over_mass_all_nominal: exact strict-ZOH lag cells



selected (0.280689716, 0.280701637] s; final smooth 0.280695525 s; data boundary=False

force_over_mass_all_nominal: parameter estimate

Case: force_over_mass_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.46022173

inertia [kg m²]:

```
[[ 0.36268776  0.00787387  0.02120761]
 [ 0.00787387  0.09286837 -0.01282338]
 [ 0.02120761 -0.01282338  0.44448575]]
```

CoG body [m]: [-0.00438903 0.00182797 0.01302964]

force effectiveness: [1.04619006 1.04619015 1.04619012 1.04619005]

scale-free J/m [m²]:

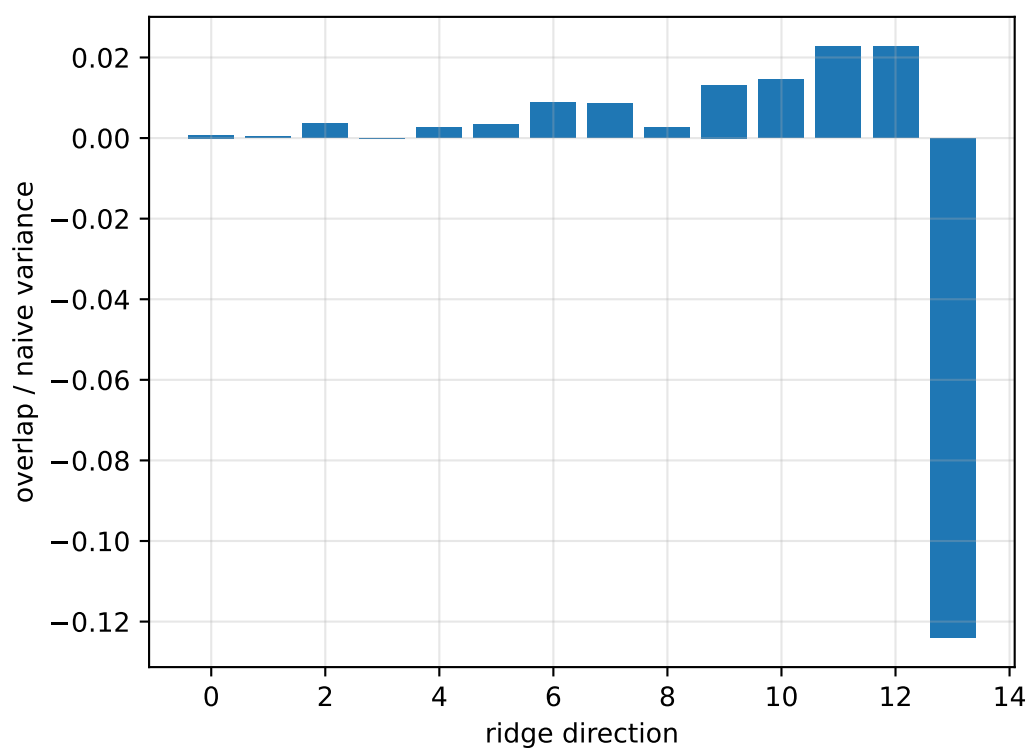
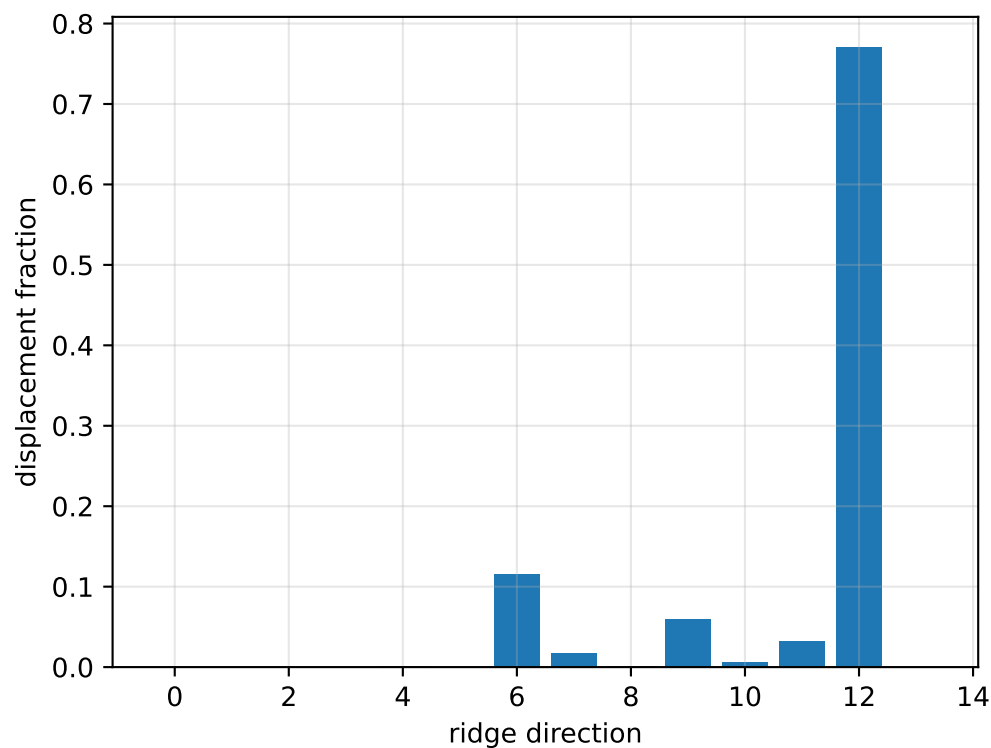
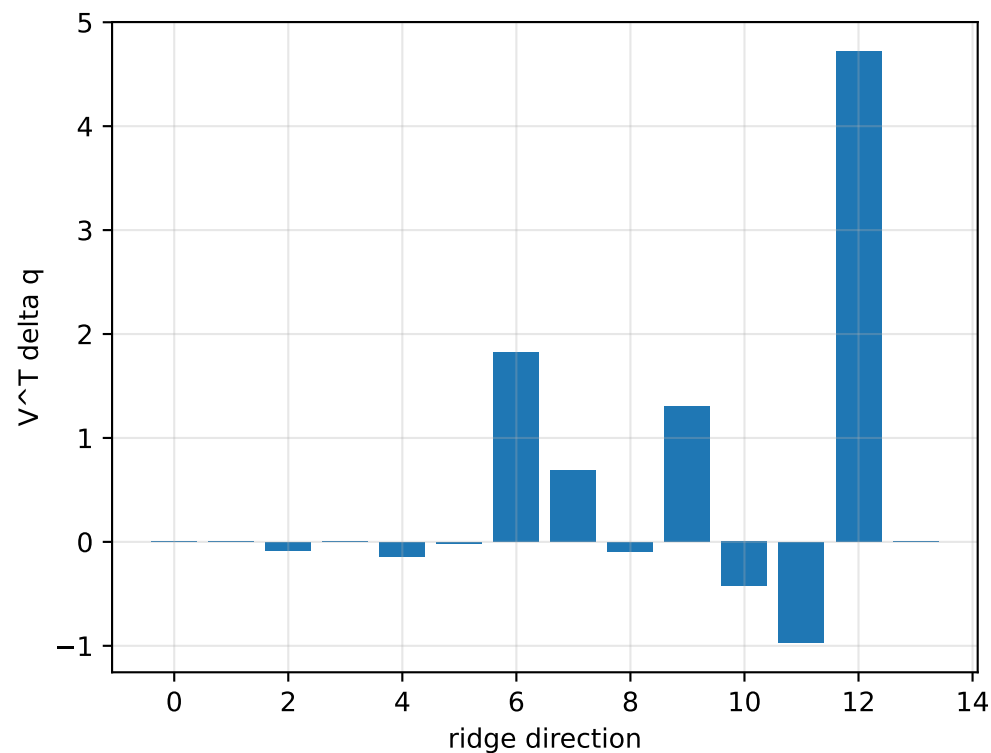
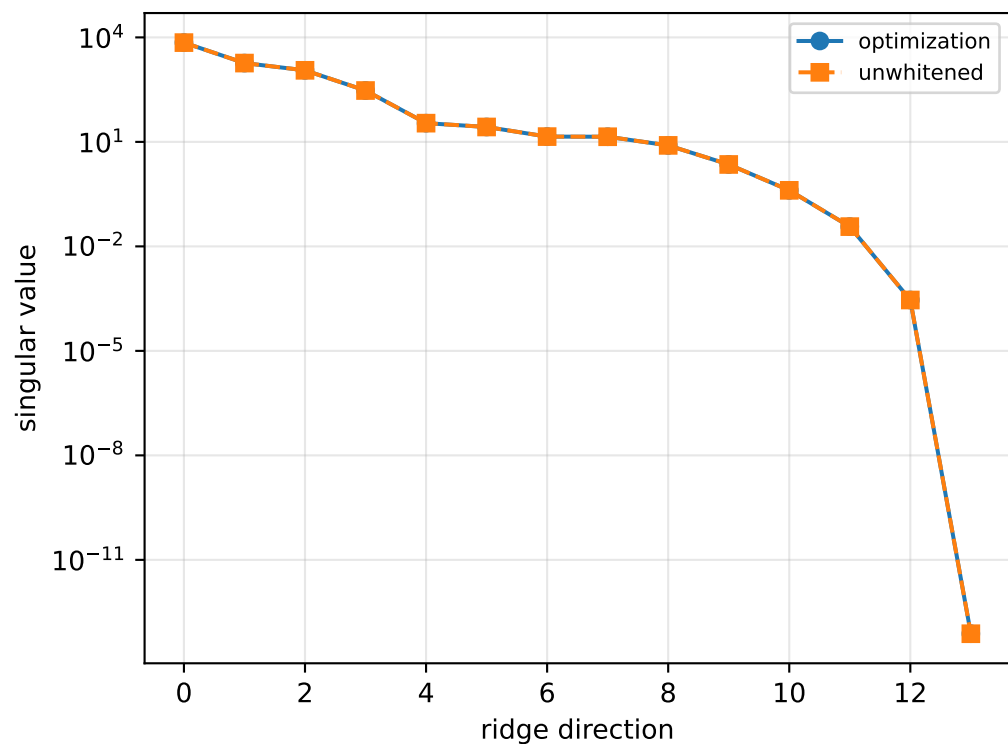
```
[[ 0.14742076  0.00320047  0.0086202 ]
 [ 0.00320047  0.03774797 -0.00521229]
 [ 0.0086202  -0.00521229  0.18066898]]
```

scale-free f/m: [0.42524218 0.42524222 0.42524221 0.42524218]

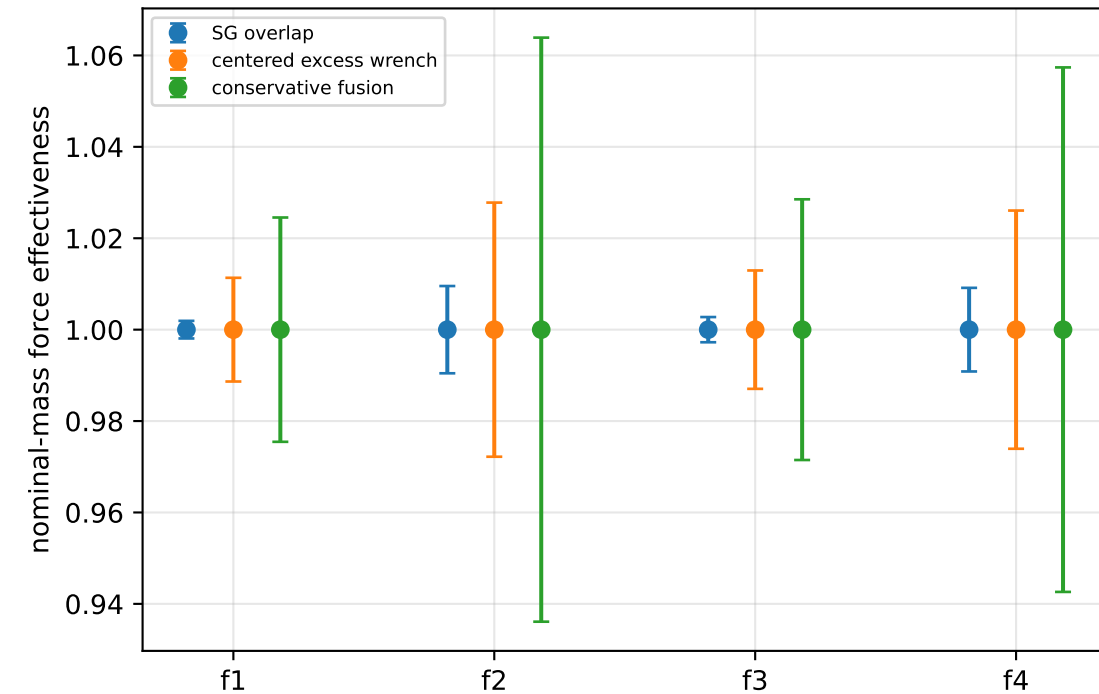
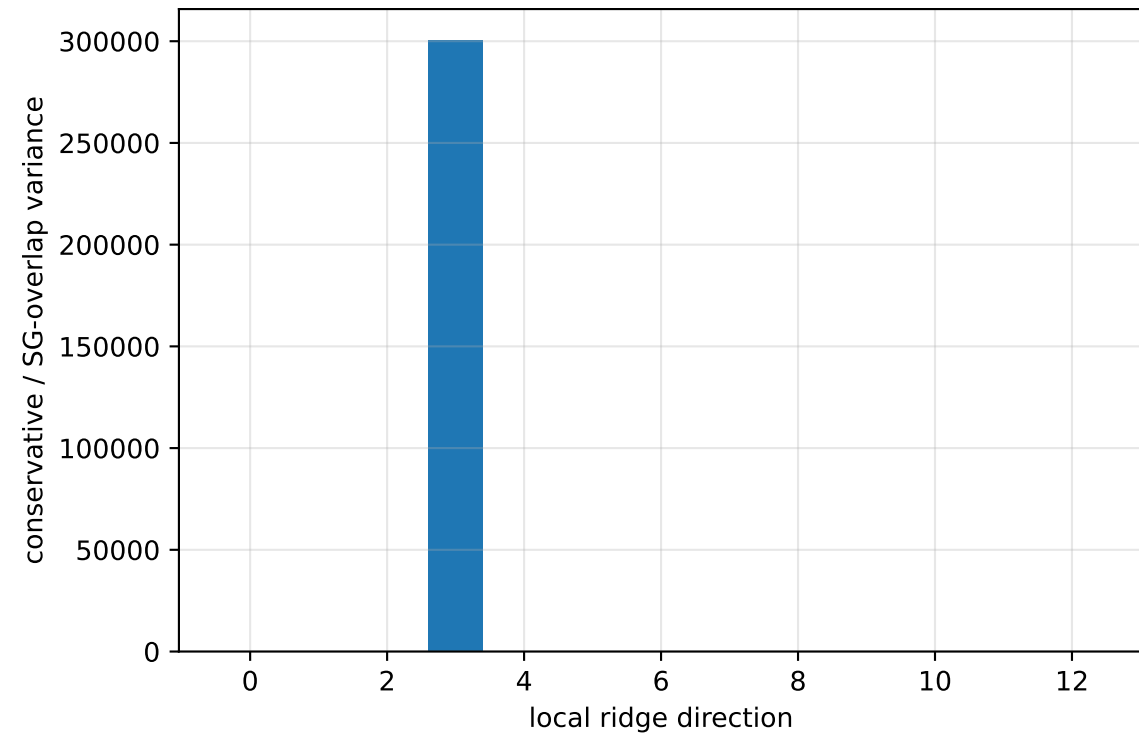
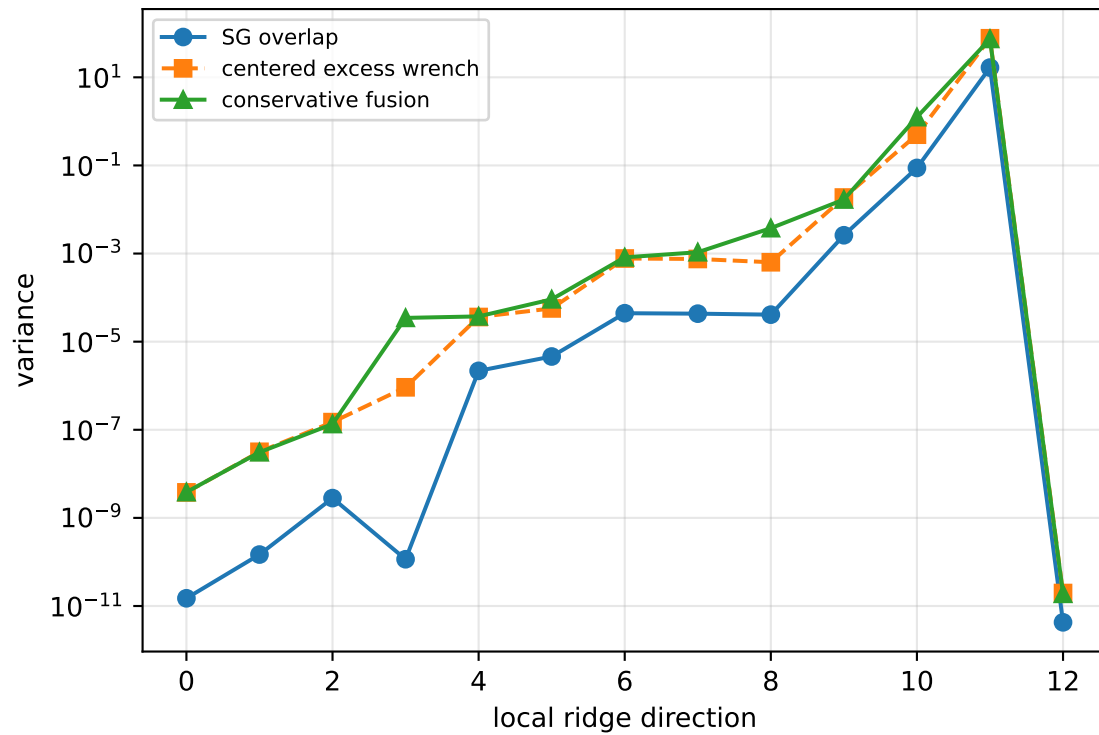
rotor lag cell: (0.280689716, 0.280701637] s

representative: 0.280695677 s

force_over_mass_all_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.37e-13)



force_over_mass_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 237.9$
wrench-to-acceleration recovery max error = $3.11\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

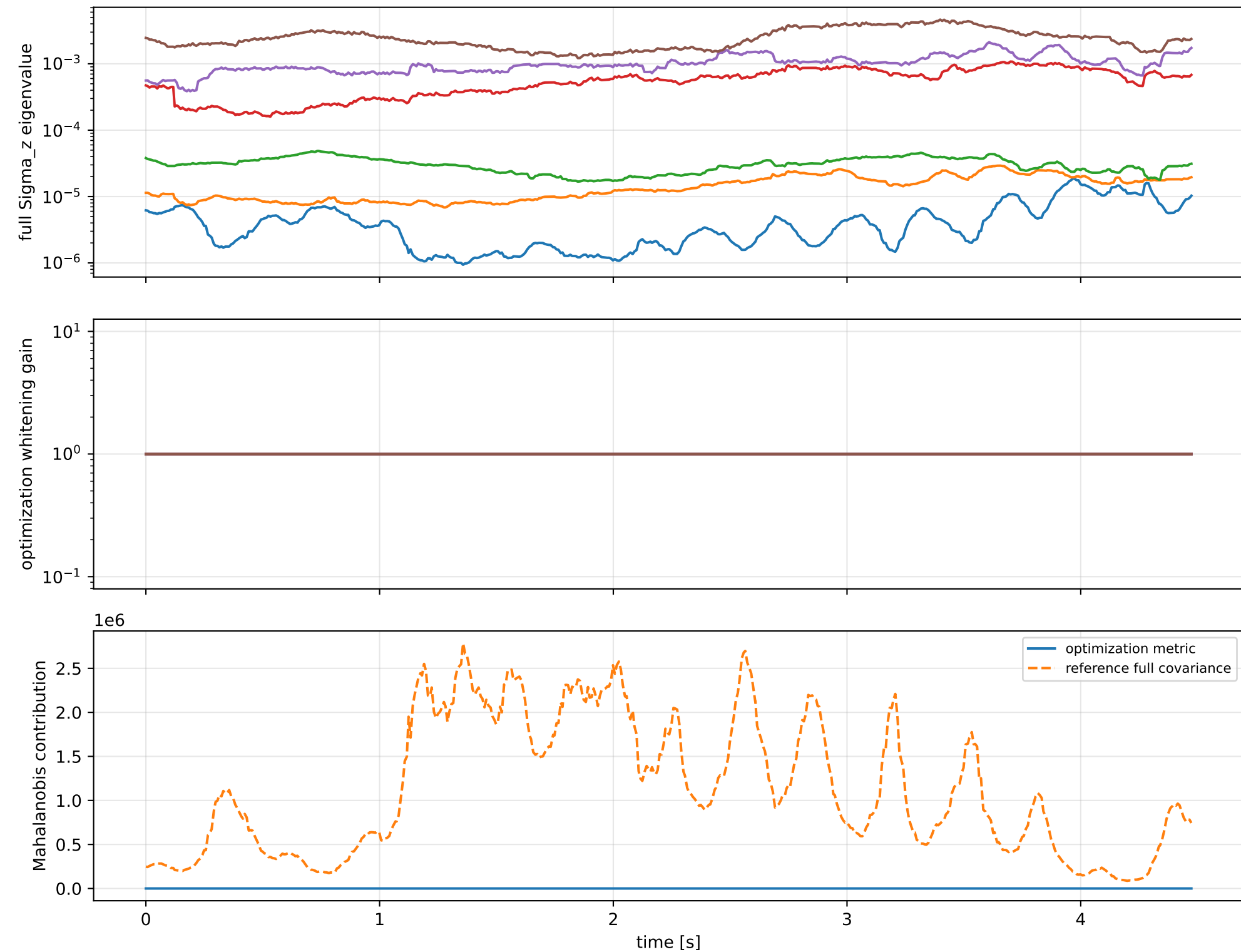
index 11: variance=74.22, ratio=4.46
 $\log_{\text{mass}}=+0.0629$; second_moment_diag_1=+0.399
second_moment_diag_2=+0.00102; second_moment_diag_3=-0.715
second_moment_offdiag_12=+0.2; second_moment_offdiag_13=-0.449
second_moment_offdiag_23=+0.259; cog_x=+3.02e-05
cog_y=-3.48e-05; cog_z=-4.98e-06
log_force_eff_1=+0.063; log_force_eff_2=+0.0632
log_force_eff_3=+0.0629; log_force_eff_4=+0.0627

index 10: variance=1.248, ratio=14.22
 $\log_{\text{mass}}=+0.0221$; second_moment_diag_1=-0.0288
second_moment_diag_2=+0.192; second_moment_diag_3=-0.275
second_moment_offdiag_12=-0.75; second_moment_offdiag_13=-0.209
second_moment_offdiag_23=-0.525; cog_x=+0.00283
cog_y=-0.00351; cog_z=+5.15e-05
log_force_eff_1=+0.0238; log_force_eff_2=+0.0521
log_force_eff_3=+0.0185; log_force_eff_4=-0.00412

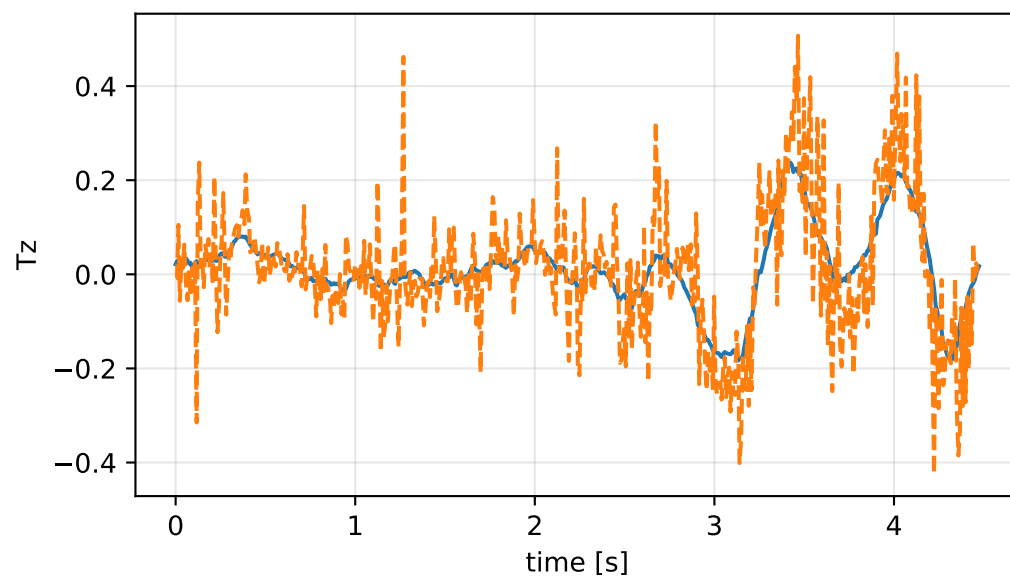
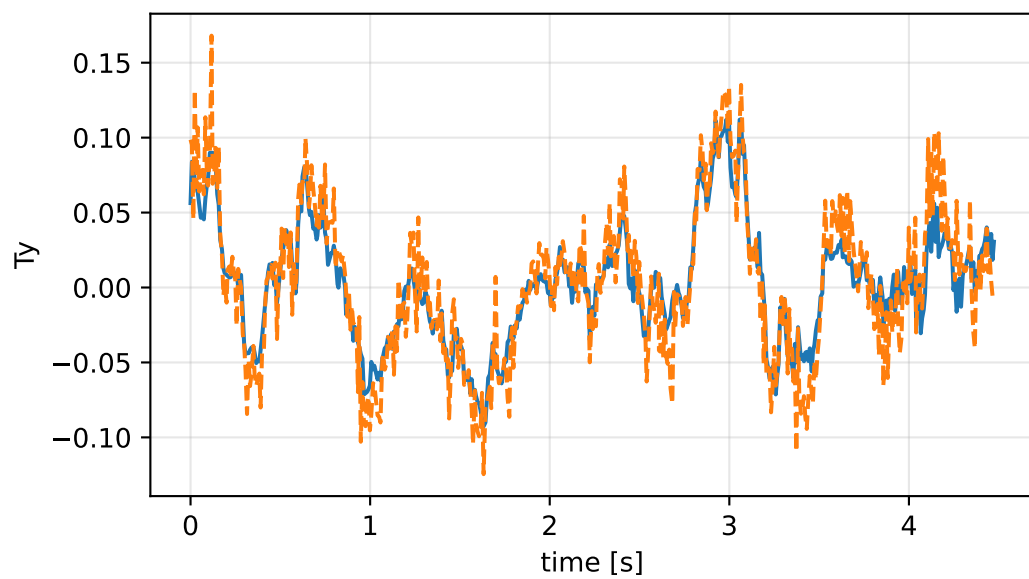
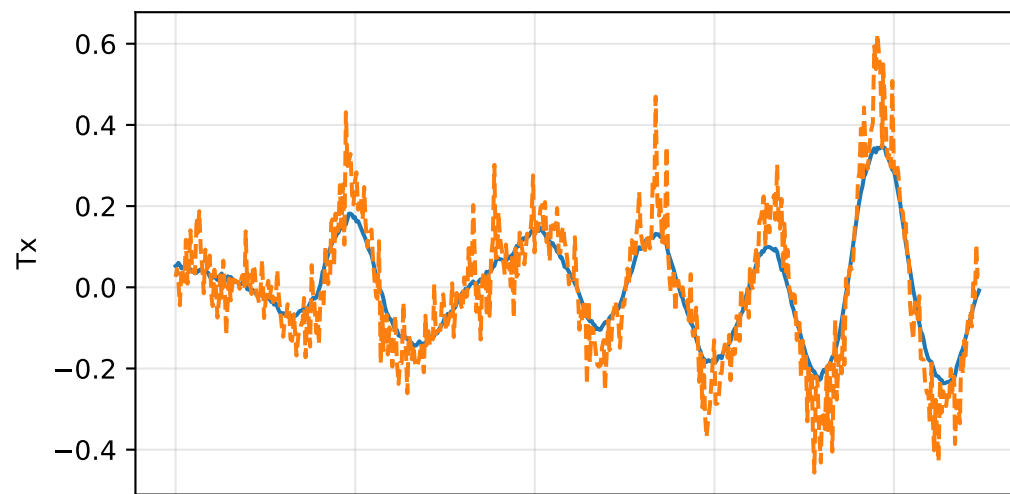
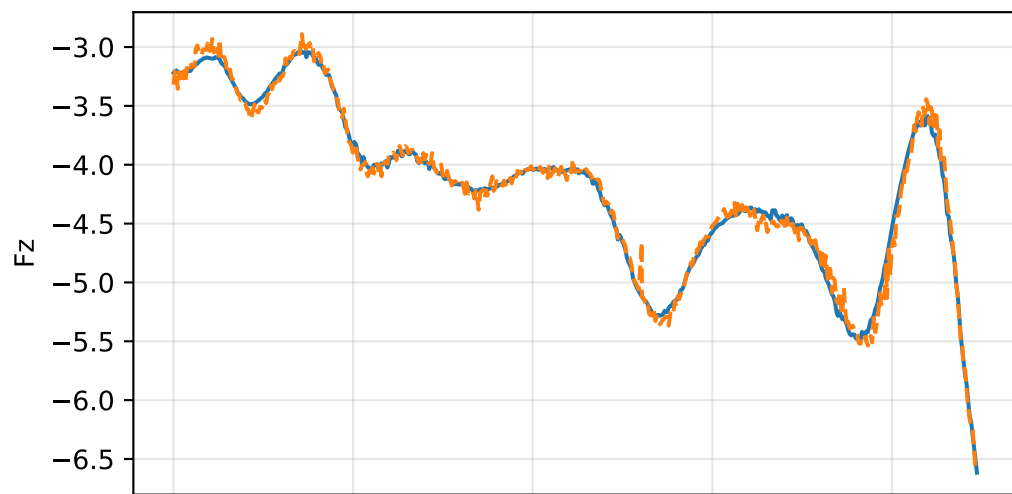
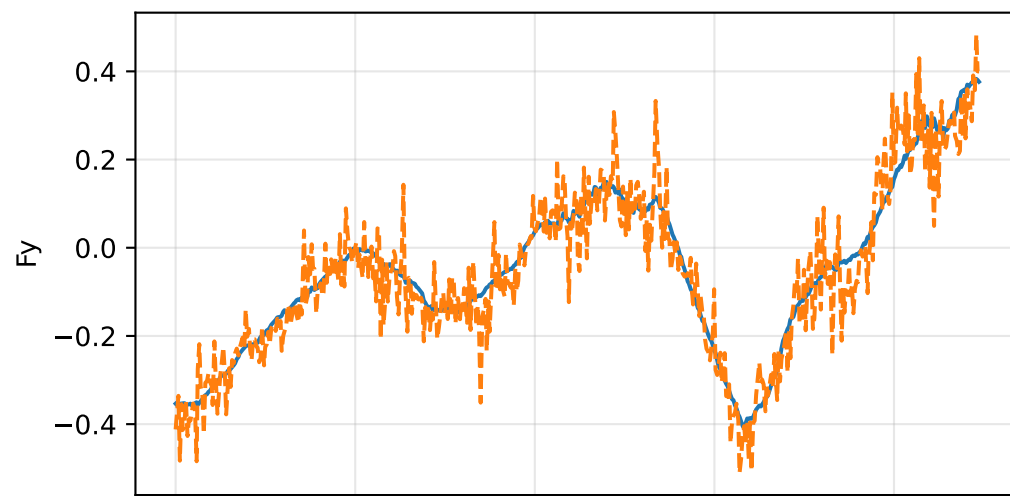
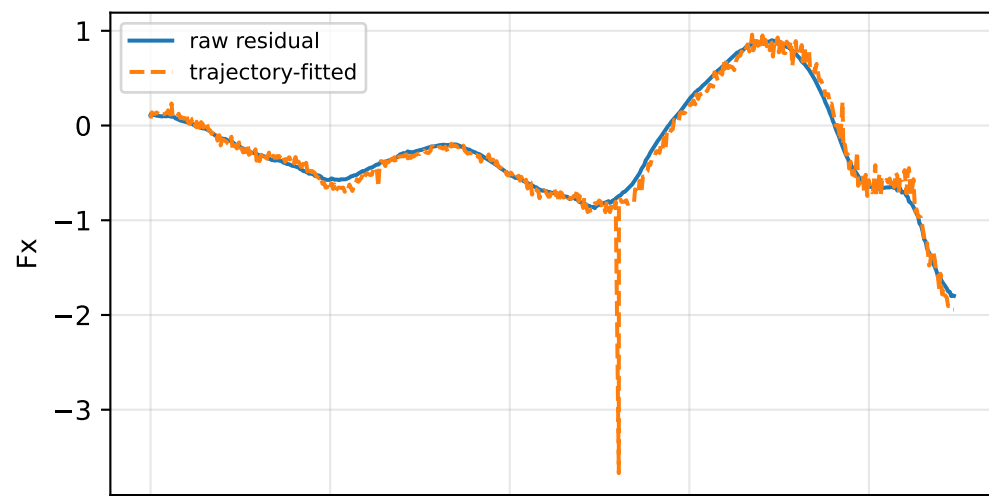
index 9: variance=0.01682, ratio=6.466
 $\log_{\text{mass}}=-0.161$; second_moment_diag_1=-0.036
second_moment_diag_2=+0.896; second_moment_diag_3=-0.0582
second_moment_offdiag_12=+0.0427; second_moment_offdiag_13=+0.1
second_moment_offdiag_23=+0.226; cog_x=+0.00432
cog_y=+0.000388; cog_z=-0.000145
log_force_eff_1=-0.181; log_force_eff_2=-0.142
log_force_eff_3=-0.139; log_force_eff_4=-0.178

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_all_nominal: optimization vs reference covariance



force_over_mass_all_nominal: wrench / closure (force max $7.11\text{e-}15$, torque max $9.71\text{e-}17$)



force_over_mass_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/force_over_mass_

SHA256: 1299af3479d15bf1e5d1f52aba937e63fa8162cc154062ba161787f233f29dd2

data objective: 913.951343302

prior objective: 0.00786994252979

total objective: 913.959213244

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_all_nominal (force_effectiveness_over_mass)

components: rotor_1, rotor_2, rotor_3, rotor_4

target: [0.42524282 0.42524282 0.42524282 0.42524282]

std: [1.e-05 1.e-05 1.e-05 1.e-05]

final: [0.42524218 0.42524222 0.42524221 0.42524218]

error: [-6.42224575e-07 -6.03203439e-07 -6.16387800e-07 -6.46334118e-07]

standardized residual: [-0.06422246 -0.06032034 -0.06163878 -0.06463341]

factor objective: 0.00786994252979